CHEMISTRY ONLY

10 ELECTROLYSIS

In this chapter we will look at what happens when we pass electricity through various substances.

Some things don't change when we pass electricity through them. For example, if you attach a piece of copper to a battery and make a complete circuit you may notice that the copper gets hot, but apart from that there is no other change. However, when you pass electricity through a solution of potassium iodide, a chemical change occurs, and hydrogen gas and iodine are formed. We will learn about why this happens in this chapter.



▶ Figure 10.1 This photograph shows what happens if you connect a solution of potassium iodide into a simple electrical circuit. If you look at what is happening in the solution, you can see obvious signs of chemical change. Some coloured substance is being produced at the positive electrode and a gas is being given off at the negative electrode.

LEARNING OBJECTIVES

- Understand why covalent compounds do not conduct electricity
- Understand why ionic compounds conduct electricity only when molten or in aqueous solution
- Know that anion and cation are terms used to refer to negative and positive ions, respectively
- Describe experiments to investigate electrolysis, using inert electrodes, of molten compounds (including lead(II) bromide) and aqueous solutions (including sodium chloride, dilute sulfuric acid and copper(II) sulfate) and to predict the products
- Write ionic half-equations representing the reactions at the electrodes during electrolysis and understand why these reactions are classified as oxidation or reduction
- Practical: Investigate the electrolysis of aqueous solutions

WHY THINGS CONDUCT ELECTRICITY

Before we continue, we need to remind ourselves about why things do or don't conduct electricity. In order for things to conduct electricity, there must be charged particles present and these charged particles must be free to move. The charged particles will be either electrons or ions; it is important that you are clear which one you are talking about.

METALS

If you remember, the structure of a metal is made up of a lattice of positive ions surrounded by a sea of delocalised electrons. Metals conduct electricity because the delocalised electrons are free to move.

IONIC COMPOUNDS

These are compounds such as sodium chloride and potassium iodide. Ionic compounds don't conduct electricity when they are solid because the ions are held tightly in position in the lattice – they are not free to move around (they can only vibrate). They do, however, conduct electricity when they are molten (have melted) or if they are dissolved in water (in aqueous solution). This happens because the ions then become free to move around.

Remember that ionic compounds are made up of positive ions and negative ions:

- anions are negative ions
- cations are positive ions.

We will look at where these names come from later in the chapter.

COVALENT COMPOUNDS

Covalent molecular compounds are substances such as hexane (C_6H_{14}), methane (CH_4) and carbon dioxide (CO_2). These do not conduct electricity in any state or in solution. Covalent molecular compounds consist of individual molecules. These molecules don't have any overall electrical charge, so there are no charged particles to move around. Also, all the electrons are held tightly in the atoms or in covalent bonds and so they are not able to move from molecule to molecule.

REMINDER

The arrow here shows that this is a reversible reaction. Reversible reactions are discussed in Chapter 21.

There are some exceptions to this, such as covalent compounds that form ions as they react with water, for example ammonia:

$$NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

Ammonia solution conducts electricity because there are ions which are free to move.

Hydrogen chloride gas dissolves in water to form hydrochloric acid (HCl(aq)). Hydrogen chloride ionises in water:

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$

PASSING ELECTRICITY THROUGH COMPOUNDS: ELECTROLYSIS

When metals conduct electricity you will not notice anything happening, except perhaps that the metal gets hotter. When you pass electricity through an ionic compound, either molten or in solution, a chemical reaction occurs.

Electrolysis is a chemical change caused by passing an electric current through a compound which is either molten or in solution.

SOME OTHER IMPORTANT WORDS

An **electrolyte** is a liquid or solution that undergoes electrolysis. Electrolytes all contain ions. The movement of the ions is responsible for both the conduction of electricity and the chemical changes that take place.

The electricity is passed into and out of the electrolyte through two **electrodes**. Carbon is frequently used for electrodes because it conducts electricity and is chemically fairly **inert** (this means that it does not react with things). Platinum is also fairly inert and can be used instead of carbon. Various other metals are sometimes used as well.

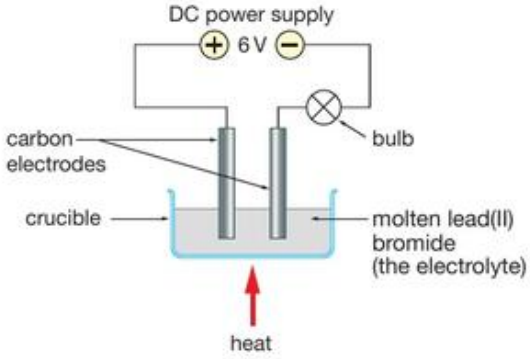
HINT

Remember **PANiC**: positive anode, negative (is) cathode.

The positive electrode is called the **anode**. The negative electrode is called the **cathode**.

THE ELECTROLYSIS OF MOLTEN COMPOUNDS

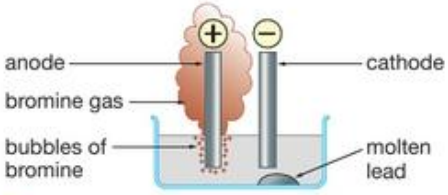
ELECTROLYSING MOLTEN LEAD(II) BROMIDE, $PbBr_2$



▲ Figure 10.2 Electrolysing molten lead(II) bromide

KEY POINT

The power supply can be a 6 volt battery or a power pack. It doesn't matter which. The voltage isn't critical either.

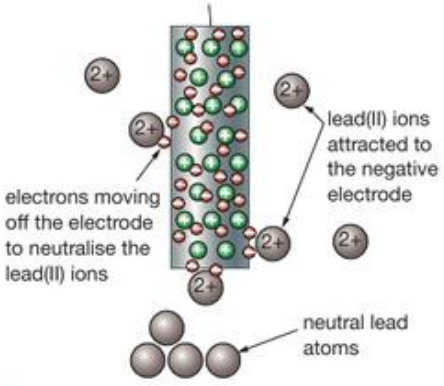


▲ Figure 10.3 What happens when the lead(II) bromide melts and electricity passes through it (the bulb also lights up).

Nothing at all happens until the lead(II) bromide melts. Then:

- the bulb lights up, showing that electrons are flowing through it
- there is bubbling around the electrode (the anode) connected to the positive terminal of the power source as brown bromine gas is given off
- nothing seems to be happening at the electrode (the cathode) connected to the negative terminal of the power source, but afterwards metallic lead is found underneath it
- when you stop heating and the lead(II) bromide solidifies again, everything stops, there is no more bubbling and the bulb goes out.

EXPLAINING WHAT IS HAPPENING



▲ Figure 10.4 The reaction at the cathode

Lead(II) bromide is an ionic compound. The solid consists of a giant structure of lead(II) ions (Pb^{2+}) and bromide ions (Br^-) packed regularly in a crystal lattice. The ions are locked tightly in the lattice and aren't free to move. The solid lead(II) bromide doesn't conduct electricity. As soon as the solid melts, the ions become free to move around.

As soon as you connect the power source in Figure 10.2, it pumps any mobile electrons away from the left-hand electrode towards the right-hand one. This means that there are extra electrons at the right-hand electrode, so it is negative. The left-hand electrode is positive because some of the electrons have been removed from it. The positive lead(II) ions in the molten lead(II) bromide are attracted to the negative electrode, the cathode. When they get there, each lead(II) ion picks up two electrons from the electrode and forms neutral lead atoms (Figure 10.4). These fall to the bottom of the container as molten lead. This can be represented by a **half-equation**.

$$Pb^{2+} + 2e^- \rightarrow Pb$$

The power source pumps new electrons along the wire to replace the electrons that have been removed from the cathode.

Bromide ions are attracted to the positive electrode, the anode (Figure 10.5). When they get to the positive electrode, the extra electron which makes the bromide ion negatively charged moves onto the electrode. This is because this electrode is short of electrons.

The loss of the extra electron converts each bromide ion into a bromine atom:

$$Br^- \rightarrow Br + e^-$$

KEY POINT

Half-equations show either oxidation or reduction reactions (see below). Electrons are shown as e^- in half-equations.

KEY POINT

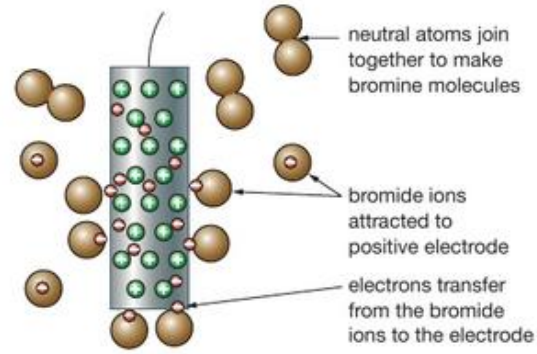
A bromine atom has only 7 electrons in its outer shell so it joins to another Br atom to form a covalent bond. Both atoms then have 8 electrons in their outer shell.

These join in pairs to make bromine molecules:

$$2Br \rightarrow Br_2$$

Overall:

$$2Br^- \rightarrow Br_2 + 2e^-$$



▲ Figure 10.5 The reaction at the anode

KEY POINT

It is really important to use the correct terms when talking about the reactions at the electrodes. Bromide ions are attracted to the anode, where they lose electrons to form bromine molecules. Do not confuse bromide and bromine.

The new electrons on the electrode flow back into the power source. Because electrons are flowing in the external circuit, the bulb lights up.

KEY POINT

The external circuit is the wire, power pack, the bulb and the electrodes. **Electrons** flow in the external circuit but **ions** flow in the electrolyte.

We sometimes talk about ions being **discharged** at the electrodes. Discharging an ion means that it loses its charge. This happens either by giving up electron(s) to the electrode or receiving electron(s) from it. We can therefore say that bromide ions and lead(II) ions were discharged at the electrodes.

ELECTROLYSIS AND REDOX

You will learn more about **oxidation** and **reduction** reactions in Chapter 14. Oxidation and reduction are words used to describe what is happening to things in certain chemical reactions. We can define oxidation and reduction as follows:

- *Oxidation occurs when something loses electrons.*
- *Reduction occurs when something gains electrons.*

HINT

OILRIG:
Oxidation
Is
Loss of electrons
Reduction
Is
Gain of electrons

We usually simply shorten this to oxidation is loss of electrons and reduction is gain of electrons. A way of remembering this is the mnemonic OILRIG.

If we look again at the electrode equations in the electrolysis of lead(II) bromide, we see that the lead(II) ions gain electrons at the cathode:

$$Pb^{2+} + 2e^- \rightarrow Pb$$

Gain of electrons is reduction. The lead(II) ions are reduced to lead atoms.

The bromide ions lose electrons at the anode:

$$2Br^- \rightarrow Br_2 + 2e^-$$

Loss of electrons is oxidation. Bromide ions are oxidised to bromine molecules.

If something loses electrons something else must gain electrons and so oxidation and reduction always occur at the same time: we talk about **redox** reactions, **red**(uction)**ox**(idation). In the reactions going on at the electrodes each equation only shows one of the processes occurring, either oxidation or reduction, and so we call these *half-equations*. Both these reactions involve ions and so, in the exam, you may be asked to write *ionic half-equations* representing the reactions at the electrodes.

THE ELECTROLYSIS OF OTHER MOLTEN SUBSTANCES

If we carry out electrolysis of molten sodium chloride, we get sodium at the cathode (negative electrode) and chlorine at the anode (positive electrode). The ionic half-equations are:

cathode: $Na^+ + e^- \rightarrow Na$ reduction

Sodium ions are reduced to sodium atoms.

anode: $2Cl^- \rightarrow Cl_2 + 2e^-$ oxidation

Chloride ions are oxidised to chlorine molecules.

These half-equations must always balance in terms of the number of atoms on each side, but also in terms of the charges; the total charge must be the same on both sides. This is why we need 2 electrons in the second half-equation but only 1 in the first.

Let us look at another example, the electrolysis of molten aluminium oxide (Al_2O_3). We get aluminium at the cathode and oxygen at the anode. Aluminium is in Group 3 in the Periodic Table and so an aluminium atom has 3 electrons in its outer shell. You may remember from Chapter 7 that this means that Al forms a 3+ ion. This 3+ ion will be attracted to the negative electrode in electrolysis and we will get the reaction:

$Al^{3+} + 3e^- \rightarrow Al$

In order to cancel out the 3+ charge on the Al^{3+} ion we need to add 3 electrons. The total charge on the left-hand side is zero (**3+** + **3–**), which is the same as on the right-hand side. Another way to think about this is that if an aluminium atom loses 3 electrons to form an Al^{3+} ion, we must put those electrons back in order to form the atom again. This is a reduction reaction because the Al^{3+} ion gains electrons.

When we electrolyse molten aluminium oxide, the half-equation for the reaction at the anode is:

$2O^{2-} \rightarrow O_2 + 4e^-$

HINT

A common mistake is to put the electrons on the wrong side of the half-equation. Check the charges to make sure you have the same total charge on both sides.

Each O^{2-} ion has 2 'extra' electrons and oxygen atoms go around in pairs, so we must remove 4 electrons to form O_2. These are shown on the right-hand side of the half-equation. There are two Os on each side and the total charge on each side is 4–. This is an oxidation reaction.

When molten zinc(II) chloride is electrolysed, zinc is obtained at the cathode and chlorine at the anode:

cathode: $Zn^{2+} + 2e^- \rightarrow Zn$ reduction

anode: $2Cl^- \rightarrow Cl_2 + 2e^-$ oxidation

We can make the following generalisations from the reactions above:

- If you electrolyse a molten ionic compound only containing two elements, you will get the metal at the cathode (because metals form positive ions) and the non-metal at the anode (because non-metals form negative ions).
- Reduction always occurs at the cathode and oxidation always occurs at the anode.

HINT

A way of remembering this is AN OX RED CAT or AN OILRIG CAT.

You can probably see now that positive ions are known as cations because they are attracted to the cathode (negative electrode). Negative ions are known as anions because they are attracted to the anode (positive electrode).

Not all ionic compounds can be electrolysed when they are molten. Some break up into simpler substances before their melting point. For example, copper(II) carbonate breaks into copper(II) oxide and carbon dioxide, even on gentle heating. It is impossible to melt it.

THE ELECTROLYSIS OF AQUEOUS SOLUTIONS

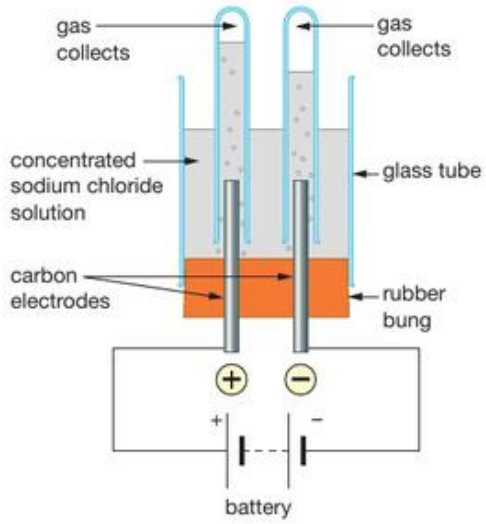
When aqueous solutions are electrolysed the products are not always the same as when molten salts are electrolysed.

ACTIVITY 5

▼ PRACTICAL: INVESTIGATING THE ELECTROLYSIS OF AQUEOUS SOLUTIONS

! Safety Note: Wear eye protection. Do not smell the chlorine, especially if you have asthma. Once chlorine is detected the current must be switched off.

We can investigate the electrolysis of an aqueous solution such as sodium chloride solution using the apparatus shown in Figure 10.6.



▲ Figure 10.6 Electrolysis of sodium chloride solution

The following procedure is used:

- Set up the apparatus as shown in Figure 10.6. The glass tube, rubber bung and electrodes together are sometimes called an **electrolytic cell**.
- Pour concentrated sodium chloride solution into the glass tube.
- Place a test-tube containing sodium chloride solution over each electrode. The test-tubes must not completely cover the electrodes or ions will be unable to flow and there will be no current.
- Connect the battery/powerpack to the electrodes.
- The experiment should be done in a fume cupboard (fume hood) or well-ventilated room because chlorine gas is poisonous.

We can see if something is happening by looking for bubbles of gas or a metal forming at the electrodes. Any gases can be tested.

The tests for gases are covered in Chapter 18.

In this experiment we see bubbles of gas at both electrodes. When the gases are tested we find that hydrogen forms at the negative electrode (cathode) and chlorine forms at the positive electrode (anode).

KEY POINT

We would expect to get the same volume of hydrogen as chlorine. However, in reality we appear to obtain less chlorine than expected because it is more soluble in water.

THE ELECTROLYSIS OF SODIUM CHLORIDE SOLUTION

When molten sodium chloride is electrolysed, the products at the electrodes are:

anode: chlorine

cathode: sodium

When you electrolyse sodium chloride solution you do not get the same products as when you electrolyse molten sodium chloride. Although chlorine is still formed at the anode, hydrogen is produced at the cathode rather than sodium. The hydrogen at the cathode comes from the water.

KEY POINT

The reversible sign shows that as water molecules break up to form hydrogen ions and hydroxide ions, these ions are recombining to make water again.

Water is called a *weak electrolyte*. It ionises very slightly to give hydrogen ions and hydroxide ions:

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

Whenever you have water present, you have to consider these ions *as well as* the ions in the compound you are electrolysing.

AT THE CATHODE

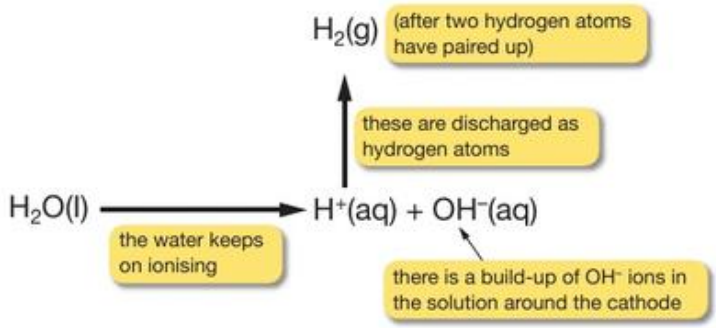
The solution contains $Na^+(aq)$ and $H^+(aq)$, and these are both attracted to the negative electrode (cathode). However, sodium is a very reactive metal. This means that it is very difficult to add an electron to a sodium ion to convert it back to a sodium atom. Hydrogen is less reactive than sodium so it is easier to add an electron to a hydrogen ion to form a hydrogen atom. Each hydrogen atom formed combines with another one to make a hydrogen molecule:

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

KEY POINT

The more reactive something is, the greater tendency it has to form an ion. This means that the more reactive something is, the more difficult it is to turn it back into an atom. For positive ions, the lower the position of an element in the reactivity series, the more easily it will accept an electron. The reactivity series is discussed in Chapter 14.

Remember that the hydrogen ions come from water molecules splitting up. Each time a water molecule ionises, it also produces a hydroxide ion. There is a build-up of these in the solution around the cathode. These hydroxide ions make the solution alkaline in the region around the cathode.



▲ Figure 10.7 The process of electrolysing solutions

KEY POINT

Because the hydrogen ions are discharged (removed from the solution as hydrogen gas), they can no longer react with the hydroxide ions and reform water. The ionisation of the water becomes a one-way process.

There is an alternative way of looking at this cathode reaction, starting from neutral water molecules, which can be thought of as taking electrons directly from the cathode:

$$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$$

HINT

You might come across either half-equation in the exam. Either should be accepted in answers.

You can see more easily why the solution becomes alkaline using this half-equation as $OH^-(aq)$ ions are produced.

AT THE ANODE

$Cl^-(aq)$ and $OH^-(aq)$ are both attracted by the positive anode. It is slightly easier to remove electrons from (oxidise) the hydroxide ion than from the chloride ion, but there isn't much difference. There are, however, many, many more chloride ions present in the solution, and so it is mainly these that are oxidised at the anode:

$$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$$

EXTENSION WORK

If the sodium chloride solution is dilute, you get noticeable amounts of oxygen produced as well as chlorine. This comes from the hydroxide ions:

$$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$$

It is unlikely that you will be asked about this at International GCSE.

THE REMAINING SOLUTION

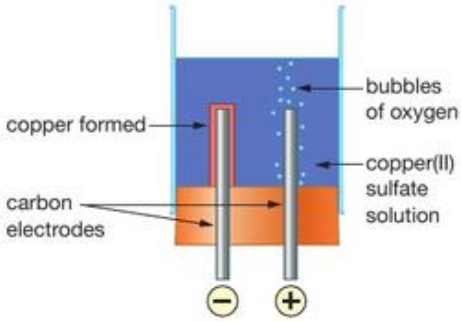
If the electrolysis is carried on for a long time, we can work out what the final solution remaining at the end will be. The ions in the solution were:

from NaCl: Na^+ ~~Cl^-~~

from H_2O: ~~H^+~~ OH^-

Cl^- and H^+ ions were removed from the solution by being discharged at the electrodes, so we are left with Na^+ and OH^-, sodium hydroxide solution.

THE ELECTROLYSIS OF COPPER(II) SULFATE SOLUTION USING INERT ELECTRODES



▲ Figure 10.8 Electrolysis of copper(II) sulfate solution using inert electrodes produces copper at the cathode and oxygen gas at the anode.

The copper(II) ions and hydrogen ions (from the water) will be attracted to the cathode. Copper is below hydrogen in the reactivity series, which means that it is easier to add electrons to copper ions to form an atom. The cathode will get coated with pink-brown copper:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s) \quad \text{reduction}$$

Sulfate ions and hydroxide ions (from the water) will be attracted to the anode. Sulfate ions aren't easy to oxidise. Instead, you get oxygen from the oxidation of hydroxide ions from the water:

$$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^- \quad \text{oxidation}$$

There is an alternative way of looking at this anode reaction. The equation this time is:

$$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$$

You can see more easily why the solution becomes acidic using this half-equation: $H^+(aq)$ ions are produced. You may come across either half-equation in the exam.

If the electrolysis is continued for a long time the copper(II) ions will eventually all be used up, and so the colour of the solution will fade from blue to colourless. What is left in the solution? The ions originally present were:

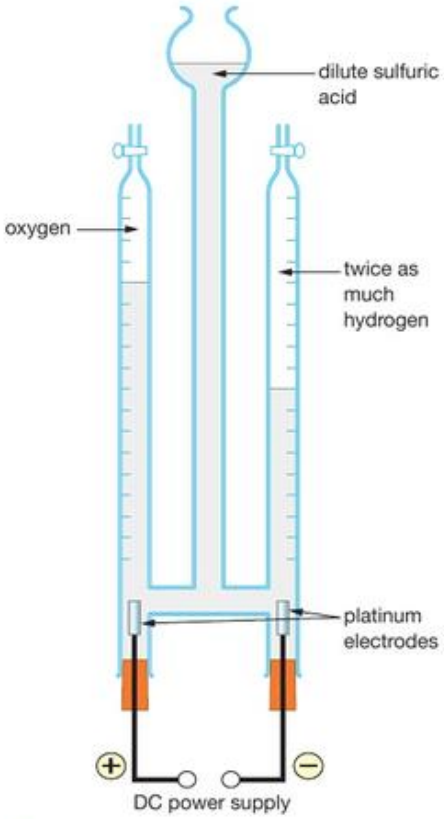
from $CuSO_4$: ~~Cu^{2+}~~ SO_4^{2-}

from water: H^+ ~~OH^-~~

Copper ions and hydroxide ions are discharged at the electrodes. Hydrogen ions from the water aren't being discharged and neither are the sulfate ions. The solution turns into dilute sulfuric acid (H_2SO_4). The electrolysis will then continue as for dilute sulfuric acid (see below).

EXTENSION WORK

The reason that we keep stressing the use of inert electrodes is that what the electrode is made of can sometimes affect the products you get. If you use copper electrodes oxygen gas is not produced at the anode but rather the anode gets smaller.



▲ Figure 10.9 Apparatus for electrolysing dilute sulfuric acid and measuring the volume of gases produced.

DID YOU KNOW?

This experiment could be used to show that the formula of water is H_2O.

THE ELECTROLYSIS OF DILUTE SULFURIC ACID USING INERT ELECTRODES

In this case, the only positive ions arriving at the cathode are hydrogen ions (from the acid and the water). These are discharged to give hydrogen gas:

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

At the anode – as with copper(II) sulfate solution – sulfate ions and hydroxide ions (from the water) arrive. The sulfate ions are too difficult to oxidise, and so you obtain oxygen from the oxidation of hydroxide ions from the water:

$$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$$

Twice as much hydrogen is produced as oxygen. Look at the half-equations above. For every 4 electrons that flow around the circuit, you would get 1 molecule of oxygen. But 4 electrons would produce 2 molecules of hydrogen. You get twice the number of molecules of hydrogen as of oxygen. Twice the number of molecules occupy twice the volume.

Actually, when we do this experiment the amount of hydrogen we obtain is more than twice as much as the oxygen. This is because oxygen is more soluble in water than hydrogen. What we sometimes do to stop this happening is to carry out the electrolysis experiment for a few minutes first in order to saturate the water with oxygen and then start to collect the gases; this gives much better results.

THE ELECTROLYSIS OF SOME OTHER SOLUTIONS USING INERT ELECTRODES

HOW TO WORK OUT WHAT WILL HAPPEN

- If the metal is high in the reactivity series, you get hydrogen produced at the cathode instead of the metal.
- If the metal is below hydrogen in the reactivity series, you obtain the metal at the cathode.
- If you have solutions of halides (chlorides, bromides or iodides), you obtain the halogen (chlorine, bromine or iodine) at the anode. With other common negative ions (sulfate, nitrate, hydroxide), you obtain oxygen at the anode.

potassium
sodium
lithium
calcium
magnesium
aluminium
(carbon)
zinc
iron
(hydrogen)
copper
silver
gold

decreasing reactivity

▲ Figure 10.10 The reactivity series shows metals in order of how reactive they are from most reactive to least reactive. This is discussed more in Chapter 14.

EXTENSION WORK

This leaves the problem of what you obtain if you have a moderately reactive metal such as zinc, for example. Reasonably concentrated solutions will give you the metal. Very dilute solutions will give you mainly hydrogen. In between, you will get both. At International GCSE you probably won't have to worry about this. The examples you will see in exams are always clear.

The table shows some simple examples of these rules.

Table 10.1 The electrolysis of solutions using inert (carbon or platinum) electrodes

Cathode (–)			Anode (+)	
Solution	Product	Half-equation	Product	Half-equation
KI(aq)	hydrogen	$2H^+(aq) + 2e^- \rightarrow H_2(g)$	iodine	$2I^-(aq) \rightarrow I_2(aq) + 2e^-$
$MgBr_2(aq)$	hydrogen	$2H^+(aq) + 2e^- \rightarrow H_2(g)$	bromine	$2Br^-(aq) \rightarrow Br_2(aq) + 2e^-$
$H_2SO_4(aq)$	hydrogen	$2H^+(aq) + 2e^- \rightarrow H_2(g)$	oxygen	$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$
$CuSO_4(aq)$	copper	$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$	oxygen	$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$

WHAT WOULD HAPPEN WITH NON-ELECTROLYTES?

For electrolysis to work, there have to be ions present. The current in the external circuit (with the bulb and power source) can flow only if there are ions which can move and be discharged.

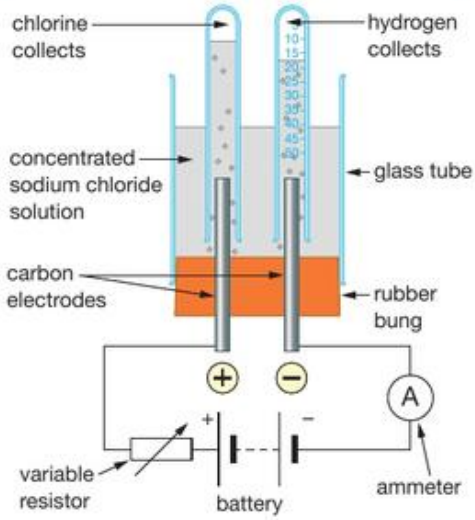
If you tried to electrolyse a covalent compound (either molten or in solution), there wouldn't be a current flow because there aren't any ions. Nothing else would happen either. Sugar, for example, is a non-electrolyte; it doesn't undergo electrolysis. It won't conduct electricity, and won't be decomposed by it, either in solution or when molten.

Simple experiments like those described in this chapter give you an easy way of finding out whether a substance is ionic or not. If it undergoes electrolysis, either molten or in solution, it must contain ions. If it doesn't undergo electrolysis, it doesn't contain ions.

KEY POINT

There are exceptions to this: covalent compounds that are electrolytes in solution. These include acids and ammonia solution.

! Safety Note: Wear eye protection. Do not smell the chlorine, especially if you have asthma. Once chlorine is detected the current must be switched off.



▲ Figure 10.11 We can use this apparatus for a quantitative electrolysis experiment.

ACTIVITY 6

▼ PRACTICAL: QUANTITATIVE ELECTROLYSIS

Quantitative means related to numbers. We can investigate how *much* product we get in an electrolysis experiment. We could, for example, investigate the effect of changing the current on the amount of hydrogen produced at the cathode in the electrolysis of sodium chloride solution. To do this we would use similar apparatus to that in Figure 10.6, but we would also have to put a variable resistor and an ammeter into the circuit to allow us to vary and measure the current. We would also have to use a gas burette or measuring cylinder to measure the volume of gas produced. We would use the following procedure:

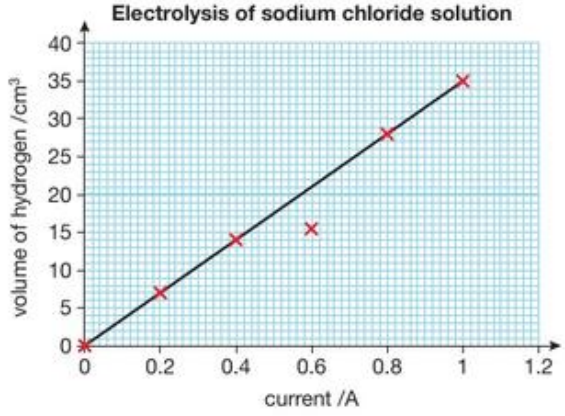
- Set up the apparatus as shown in Figure 10.11.
- Pour 50 cm^3 of concentrated sodium chloride solution into the glass tube.
- Place a gas burette (or measuring cylinder) filled with sodium chloride solution over the cathode.
- Turn on the powerpack/connect the battery and set the current to 0.2 A using the variable resistor.

- Take an initial reading on the gas burette. (We have to start the current flowing so that we can see what the current is, therefore we only take a reading on the gas burette after we have turned on the power.)
- Start the timer.
- Stop the timer after 5 minutes and note the final reading on the gas burette.
- Repeat for currents of 0.4 A, 0.6 A, 0.8 A and 1.0 A.
- Repeat each experiment to get more reliable results.

A set of results for this experiment could be:

Current/A	Volume of hydrogen gas produced/cm^3
0.20	7.0
0.40	13.9
0.60	15.1
0.80	28.0
1.0	34.9

We can plot this data on a graph (Figure 10.12).



▲ Figure 10.12 How changing the current affects the volume of hydrogen produced at the cathode.

Because the current is continuous data, we draw a line of best fit through the points. The reading at 0.6 A is an anomalous point and we do not include this when drawing our line of best fit. The reading at 0.6 A is too low and could have occurred because the current that we used was too low or some gas escaped.

The line of best fit is a straight line that goes through the origin and so we can say that we have a proportional relationship: the volume of gas produced is directly proportional to the current passed through the solution.

CHAPTER QUESTIONS

SKILLS REASONING 7

1 State what is formed at the cathode and at the anode during the electrolysis of the following substances. Assume that carbon electrodes were used each time.

a Molten lead(II) bromide

b Molten zinc chloride

c Sodium iodide solution

d Molten sodium iodide

e Copper(II) chloride solution

f Dilute hydrochloric acid

g Magnesium sulfate solution

h Sodium hydroxide solution

SKILLS PROBLEM SOLVING

2 Copy and complete the following half-equations for reaction at the anode or cathode, and state whether each involves oxidation or reduction. Electrons have been put in for the first four but not for the ones after that.

a $Mg^{2+} + e^- \rightarrow Mg$

b $Al^{3+} + e^- \rightarrow Al$

c $2Br^- \rightarrow Br_2 + e^-$

d $O^{2-} \rightarrow O_2 + e^-$

e $Cl^- \rightarrow Cl_2$

f $Ni^{2+} \rightarrow Ni$

g $OH^- \rightarrow O_2 + H_2O$

h $H_2O \rightarrow O_2 + H^+$

i $H_2O \rightarrow H_2 + OH^-$

SKILLS REASONING

3 Some solid potassium iodide was placed in an evaporating basin. Two carbon electrodes were inserted and connected to a 12 volt DC power source and a light bulb. The potassium iodide was heated. As soon as the potassium iodide was molten, the bulb came on. Purple fumes were seen coming from the positive electrode, and lilac flashes were seen around the negative one.

a Explain why the bulb didn't come on until the potassium iodide melted.

SKILLS CRITICAL THINKING 6

b State the name of the positive electrode.

8

c Name the purple fumes seen at the positive electrode, and write the ionic half-equation for their formation.

d The lilac flashes seen around the negative electrode are caused by the potassium which is formed. The potassium burns with a lilac flame. Write the ionic half-equation for the formation of the potassium.

6

e State the products formed at the electrodes if molten sodium bromide is electrolysed instead of molten potassium iodide.

8

f Write the ionic half-equations for the reactions occurring during the electrolysis of molten sodium bromide.

SKILLS CRITICAL THINKING

9

4 For electrolysis of each of the following

i write the ionic half-equation for the reaction occurring at the cathode

ii write the ionic half-equation for the reaction occurring at the anode

iii state what has been oxidised and what has been reduced.

a Molten lead(II) bromide using carbon electrodes

b Sodium chloride solution using carbon electrodes

c Calcium bromide solution using carbon electrodes

d Copper(II) sulfate solution using platinum electrodes

e Aluminium nitrate solution using carbon electrodes

f Molten magnesium iodide using carbon electrodes

g Dilute hydrochloric acid using platinum electrodes

SKILLS INTERPRETATION, REASONING

8

5 You are asked to find out whether two compounds, S and T, are electrolytes or non-electrolytes. S melts at 1261 °C and is soluble in water. T melts at 265 °C and is insoluble in water. Describe, with the aid of diagrams, how you would find out if each of these substances was an electrolyte or a non-electrolyte. In each case say what you would look for to help you to decide.

6 When copper(II) sulfate solution is electrolysed using copper electrodes the reaction at the cathode is the same as with inert electrodes but no oxygen is given off at the cathode. Instead the anode gets smaller as copper ions go into solution. The half-equation for the reaction at the anode is:

$$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$$

a Write the half-equation for the reaction at the cathode.

SKILLS REASONING

b When copper(II) sulfate solution is electrolysed using inert electrodes the blue colour of the solution fades and the solution becomes more acidic.

i Explain these observations.

9

ii Predict and explain what happens to the colour and acidity of the solution when copper(II) sulfate solution is electrolysed with copper electrodes.

END OF CHEMISTRY ONLY